

A Real-Time Data-Driven Mentorship Ecosystem for Student Performance Monitoring Using Machine Learning

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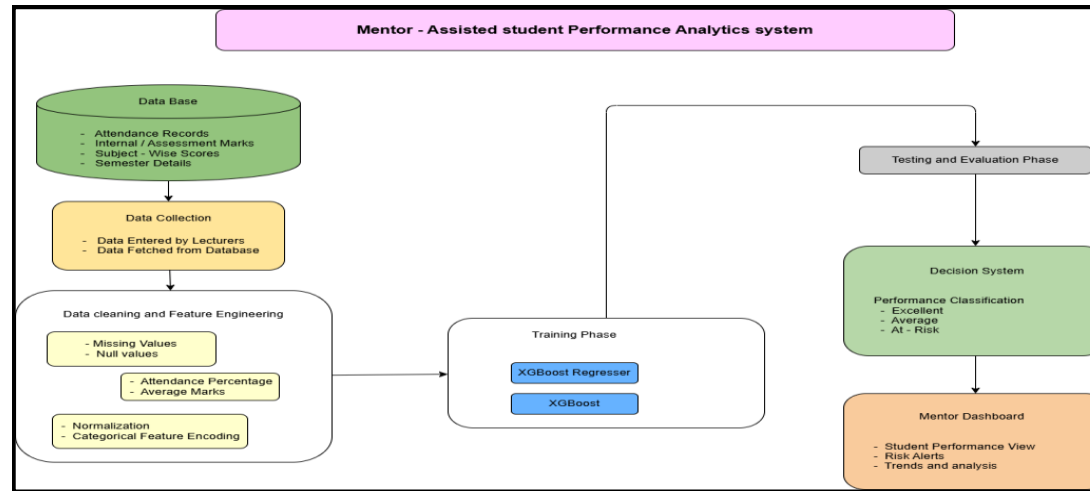
Introduction

Modern education systems generate large volumes of student data, making traditional monitoring less effective. Mentors often detect issues only after performance declines, causing delayed intervention. The proposed system uses a real-time, data-driven approach by integrating attendance and marks, while machine learning analyzes performance patterns. It provides insights and alerts to help mentors identify at-risk students early, enabling timely guidance and improving overall academic performance.

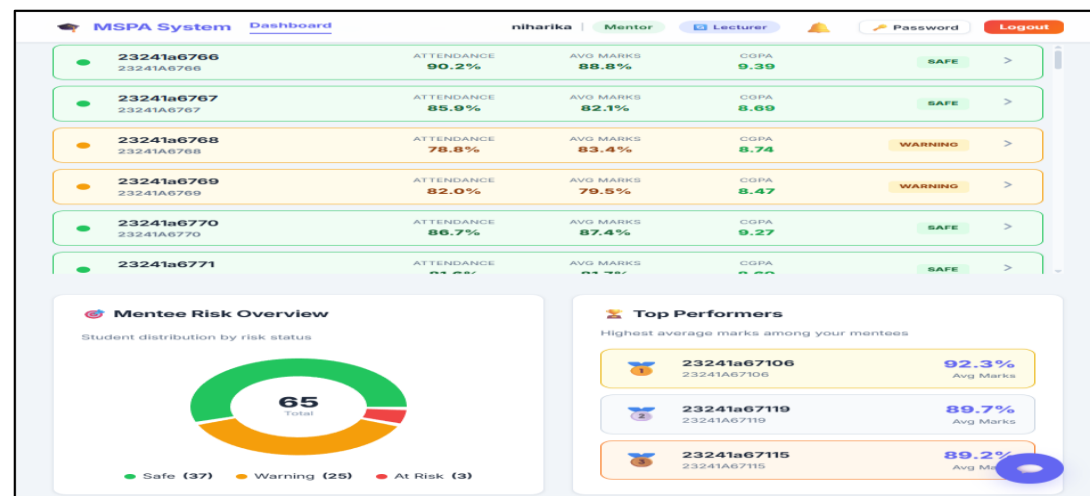
Objectives

- Uses a centralized database to store and preprocess student data
- Applies machine learning models to analyze patterns and predict performance.
- Classifies students into categories like excellent, average, and at-risk.

Architecture / Methodology



Results



Conclusion

The proposed system provides an efficient, data-driven solution for monitoring student performance by integrating academic data and applying machine learning techniques. It enables early identification of at-risk students and supports mentors with timely insights and alerts. By facilitating proactive intervention and personalized guidance, the system helps improve student outcomes and enhances the overall effectiveness of the mentoring process.

Future Work

- Integrate advanced AI techniques like deep learning and NLP to analyze student feedback and improve prediction accuracy.
- Develop a mobile application for easy access and real-time notifications for students and mentors.
- Incorporate behavioral data analysis such as engagement and activity patterns for better insights

Sustainability Goals

- **SDG 4** – Quality Education
- **SDG 9** – Industry, Innovation and Infrastructure